$\boxed{N3}$

$$\tilde{\theta}_1 = \bar{x} = \frac{1}{n}\sum_i x_i$$

а) Несмещённость

1) $\tilde{\theta}_1 = \bar{x}$

$$M\xi = \int_{-\infty}^{\infty} x\, df(x) = \int_0^{+\infty} \frac{x}{\theta} \cdot e^{-\frac{x}{\theta}} dx = \ldots$$

$$\ldots = \theta$$

$$M\tilde{\theta}_1 = M\left(\frac{1}{n}\sum_i x_i\right) = \frac{1}{n} \cdot n \cdot M\xi =$$

$$= M\xi = \theta$$

$\Longrightarrow$ несмещённая

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2 \cdot df(x) = \frac{1}{\theta}\int_0^{+\infty} x^2 \cdot e^{-\frac{x}{\theta}} dx =$$

$$= \ldots = 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

2) $\tilde{\theta}_2 = X_{(1)}$

$$\xi_{(x)} \sim \sum_{i=k}^{n} C_n^i f^i(t)(1-f(t))^{n-i}$$

$$\xi_{(x)} \sim C_3^2 (f(x))^2 (1-f(x)) + C_3^3 (f(x))^3 =$$
$$= 3(f(x))^2(1-f(x)) + (f(x))^3 =$$
$$= 3(f(x))^2 - 2(f(x))^3 = \psi(x)$$

$$p(x) = \frac{e^{-\frac{x}{\theta}}}{\theta}$$

$$f(x) = \int_0^x p(t)dt = \left(-e^{-\frac{x}{\theta}}\right)\Big|_0^x =$$
$$= (1-e^{-\frac{x}{\theta}})$$

$$\psi'(x) = 6f(x)\cdot p(x) - 6(f(x))^2 \cdot p(x) = \varphi(x)$$

$$\varphi(x) = 6(1-e^{-\frac{x}{\theta}})\frac{e^{-\frac{x}{\theta}}}{\theta} -$$
$$- 6(1-e^{-\frac{x}{\theta}})\cdot\frac{e^{-\frac{x}{\theta}}}{\theta} =$$
$$= 6(1-e^{-\frac{x}{\theta}})\cdot\frac{e^{-\frac{x}{\theta}}}{\theta}(1-1+e^{-\frac{x}{\theta}}) =$$

$$= \frac{6}{\theta}\left(e^{\frac{-2x}{\theta}} - e^{\frac{-3x}{\theta}}\right)$$

$$M\tilde{\theta}_2 = Mx_{(2)} = \int_{-\infty}^{\infty} \frac{6x}{\theta}\left(e^{\frac{-2x}{\theta}} - e^{\frac{-3x}{\theta}}\right)dx =$$

$$= 6\int_{-\infty}^{\infty} x \cdot \frac{e^{\frac{-2x}{\theta}}}{\theta} \cdot dx - 6\int_{-\infty}^{+\infty} x \cdot \frac{e^{\frac{-3x}{\theta}}}{\theta} dx =$$

$$= \ldots = \frac{5}{6}\theta \Longrightarrow \text{смещённая}$$

$$\tilde{\theta}_2' = \frac{6}{5}\tilde{\theta}_2 \Longrightarrow M\tilde{\theta}_2' = \theta \text{ (несмещ.)}$$

б) Сравнение эффективности

1) $\tilde{\theta}_1$: $D\tilde{\theta}_1 = D\left(\frac{1}{n}\sum_1 x_i\right) =$

$$= \frac{D\xi}{n} = \frac{\theta^2}{n} = \frac{\theta^2}{3}$$

2) $$M(\tilde{\theta}_2)^2 = 6\int_0^{+\infty} x^2 \cdot \frac{e^{\frac{-2x}{\theta}}}{\theta} - 6\int_0^{+\infty} x^2 \cdot \frac{e^{\frac{-3x}{\theta}}}{\theta} =$$

$$= \ldots = \frac{19}{18}\theta^2 = M(\tilde{\theta}_2)^2$$

$$D\tilde{\theta}_2 = M(\tilde{\theta}_2)^2 - (M\tilde{\theta}_2)^2 = \ldots = \frac{13}{36}\theta^2$$

$$D\tilde{\theta}_2^{\,1} = \frac{36}{25} D\tilde{\theta}_2 = \frac{13}{25}\theta^2$$

$$D\tilde{\theta}_1 = \frac{\theta^2}{5}$$

$$D\tilde{\theta}_2^{\,1} = \frac{13}{25}\theta^2$$

$$\forall \theta \in \Theta \subset (0, +\infty)$$

$$D\tilde{\theta}_1 < D\tilde{\theta}_2^{\,1}$$

$\Longrightarrow$ $\tilde{\theta}_1$ эффективное $\tilde{\theta}_2$

c) нер-во Крамера-Рао

усл. ① Регулярность модели
② Регулярность оценки

①: $f(x, \theta)$, $\theta \in \Theta \subset R$;

$f(x, \theta) = (1 - e^{-\frac{x}{\theta}}) \{(0, +\infty)\}$

– непр. дифф. на $\theta \in \Theta \subset R^3$

$$\frac{\partial}{\partial\theta}\int_{-\infty}^{+\infty} d f(x, \theta) = \int_{-\infty}^{+\infty} \frac{\partial f(x, \theta)}{\partial\theta} \Longrightarrow$$

$$\Longrightarrow \int_{A(\theta)} \frac{\partial p(x, \theta)}{\partial\theta} dx = 0$$

$$\int_0^{+\infty} \frac{\partial\left(\frac{e^{-\frac{x}{\theta}}}{\theta}\right)}{\partial\theta} dx = \ldots = 0$$

Инф-я Фишера:

$$I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2\right] =$$

$$= \ldots = \frac{1}{\theta^2} > 0$$

непр. $\forall \theta \in \Theta \subset R \Rightarrow$

$\Rightarrow$ модель регулярна

② а) $\tilde{\theta}_1$ — несмещ., модель рег.

$D\tilde{\theta}_1 = \frac{\theta^2}{3}$ — огр. на $\forall$ компакте, $\theta \in \Theta$

$\Rightarrow$ по дост. усл. оценка регулярная

б) $\tilde{\theta}_2'$ — несмещ., модель рег.,

$D\tilde{\theta}_2' = \frac{13}{25}\theta^2$ — огр. на $\forall$ компакте из $\theta \in \Theta$

$\Rightarrow \tilde{\theta}_2'$ регулярная

Нер-во Крамера-Рао работает для $\tilde{\theta}_1$, $\tilde{\theta}_2$

$$D\tilde{\theta}_1 = \frac{\theta^2}{3} = \frac{1}{n \cdot \frac{1}{\theta^2}} = \frac{\theta^2}{5}$$

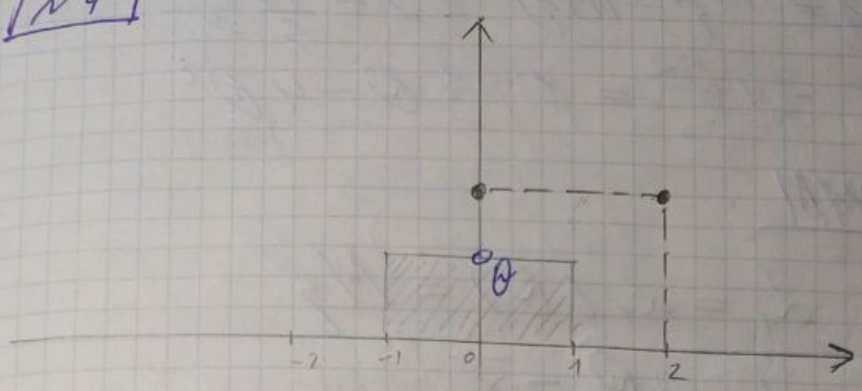
$\Rightarrow$ по дост. усл. оценка эффективна

$$D\tilde{\theta}_2 = \frac{13}{25}\theta^2 > \frac{\theta^2}{3} \rightarrow$$ ничего не знаем

По теореме о единственности эфф. оценки

$\tilde{\theta}_2$ – не эфф-я оценка.

$\boxed{№4}$




$$\xi \sim \rho(x) = \theta\{(-1,1)\setminus\{0\}\} + \left(\frac{1}{2}-\theta\right)\{0\} + \left(\frac{1}{2}-\theta\right)\{2\},$$

$$\theta \in \left(0, \frac{1}{2}\right)$$

a) $M\xi = \int_{-\infty}^{+\infty} x\,dF(x,\theta) = \int_{-1}^{1} \theta x\,dx + \left(\frac{1}{2}-\theta\right)\cdot 0 + \left(\frac{1}{2}-\theta\right)\cdot 2 = 1-2\theta$

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2\,dF(x,\theta) = \int_{-1}^{1} \theta x^2\,dx + \left(\frac{1}{2}-\theta\right)\cdot 0 + \left(\frac{1}{2}-\theta\right)\cdot 4 = 2 - \frac{10}{3}\theta$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2 - \frac{10}{5}\theta - 1 + 4\theta -$$
$$- 4\theta^2 = 1 + \frac{2}{5}\theta - 4\theta^2$$

<u>ОММ</u>.

$$\alpha_1 = \alpha_1(\theta) = M\xi$$
$$1 - 2\theta = \bar{x}$$

$$\tilde{\theta}_1 = \frac{1-\bar{x}}{2}$$

$$M\tilde{\theta}_1 = \theta : M\tilde{\theta}_1 = M\left(\frac{1-\bar{x}}{2}\right) =$$

$$= \theta \Rightarrow \theta_1 \text{ — несмещ.}$$

$$D\tilde{\theta}_1 = D\left(\frac{1-\bar{x}}{2}\right) = \frac{1}{4} D\bar{x} = \frac{1}{4}\frac{D\xi}{n} =$$

$$= \frac{1 + \frac{2}{5}\theta - 4\theta^2}{4n} \xrightarrow[n\to\infty]{} 0 \Rightarrow$$

$\Rightarrow$ по дост. усл. $\tilde{\theta}_1$ — состоят.

ОМП

$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta)$ – ф-ия правдоподобия

m чисел $\{0, 2\}$

n-m: $(-1,0) \cup (0,1)$

$$L(\theta) = \theta^{n-m}\left(\frac{1}{2} - \theta\right)^m$$

$$\ln L(\theta) = (n-m)\ln\theta + m\ln\left(\frac{1}{2} - \theta\right)$$

$$(\ln L)' = \frac{n-m}{\theta} + \frac{m}{\frac{1}{2} - \theta} \Rightarrow$$

$$\Rightarrow (n-m)\left(\frac{1}{2} - \theta\right) - m\theta = 0$$

$$(\ln L)'' = \frac{-(n-m)}{\theta^2} - \frac{m}{\left(\frac{1}{2} - \theta\right)^2} < 0$$

$\Rightarrow$ т. max

$$\frac{1}{2}n - n\theta - \frac{1}{2}m = 0$$

$$\tilde{\theta}_2 = \frac{n-m}{2n} = \frac{1}{2} - \frac{\nu}{2}; \quad \nu = \frac{m}{n}$$

в) Исслед. на несмещ., сост.

1) $M\tilde{\theta}_2 = M\left[\frac{1}{2} - \frac{1}{2}\nu\right] =$

$= \frac{1}{2} - \frac{1}{2}(1-2\theta) = \theta$

$\Rightarrow \tilde{\theta}_2$ – несмещ.

2) $D\tilde{\theta}_2 = \frac{1}{4}D\nu = \frac{p(1-p)}{n} =$

$= \frac{(1-2\theta)2\theta}{4n} \xrightarrow[n\to\infty]{} 0$

по дост. усл.

$\Longrightarrow$ состоят.

г) Исл. на эффективность

1) Регулярность модели

$p(x,\theta) = \theta\{(-1,1)\setminus\{0\}\} +$

$+ \left(\frac{1}{2}-\theta\right)\{0\} + \left(\frac{1}{2}-\theta\right)\{2\}$

– непр. дифф. по $\theta \in \left(0, \frac{1}{2}\right)$

$$\int_{-1}^{1} \frac{\partial \rho(x,\theta)}{\partial \theta} + \sum p_k(\theta) = 0$$

$$\int_{-1}^{1} dx + (-2) = 0 \quad \text{верно}$$

$$I(\theta) = M\left[\left(\frac{\partial \ln \rho(x)}{\partial \theta}\right)^2\right] =$$

$$= \ldots = \frac{2}{\theta(1-2\theta)}$$

$\Rightarrow I(\theta) > 0$ и непр. на $\theta \in (0, \frac{1}{2})$

$\Rightarrow$ модель регулярна

2) Регулярность оценки

$\tilde{\theta}_1$ – несмещ., модель регулярна

$$D\tilde{\theta}_1 = \frac{1 + \frac{2}{3}\theta + 4\theta^2}{4n}$$

– огр. на $\forall$ компакт. из $\theta \in \Theta : (0, \frac{1}{2})$

$\hat{\theta}_2$ — несмещ., модель рег.

$$D\hat{\theta}_2 = \frac{(1-2\theta)2\theta}{4}$$

— огр. на $\forall$ колебание

из $\theta \in \Theta : (0, \frac{1}{2})$

$\Rightarrow$ оценка регулярна

$\Rightarrow$ Нер-во К-Р работает для $\hat{\theta}_1$, $\hat{\theta}_2$

$$\hat{\theta}_1 : \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n} > \frac{\theta(1-2\theta)}{2n}$$

$$1 + \frac{2}{3}\theta - 4\theta^2 > 2\theta - 4\theta^2$$

$$1 + \frac{2}{3}\theta > 2\theta$$

$$1 - \frac{4}{3}\theta > 0$$

$$\theta \in (0, \frac{1}{2})$$

$\Rightarrow$ ничего сказать нельзя

$\hat{\theta}_2$ :

$\hat{\theta}_2 : \frac{(1-2\theta)2\theta}{44} = \frac{\theta(1-2\theta)}{29}$

$\Rightarrow$ по дост. усл. $\hat{\theta}_2$ – эффектив.

$\Rightarrow$ из теоремы о един-ти эфф. оценки

$\hat{\theta}_1$ – не эффектив.